

Ashutosh Swain

[GitHub](#) | [LinkedIn](#) | [Portfolio](#) | [Email](#) | [+91 97772 43905](#)

SUMMARY

Software engineer focused on backend systems in Go, building stream processing engines, infrastructure tooling, and distributed systems from scratch.

EXPERIENCE

Commenda — Software Engineering Intern 2026

Worked across backend systems and developer tooling with a focus on service reliability, observability, and production engineering.

PROJECTS

Weibo — Stream Processing Engine Repo

Built a stream processing engine in Go inspired by Apache Flink with Kafka, checkpointing, stateful operators, fault recovery, and job monitoring.

Casper — MCP Server for Terraform Infrastructure Demo | Repo

Built an MCP server for querying Terraform infrastructure with AWS drift detection, policy gates, and blast-radius simulation.

Maxim — TUI Client for Local Databases Repo

Built a terminal UI for PostgreSQL using Bubble Tea with schema-aware SQL autocomplete, credential management, and Docker-based local database management. Achieved **144 GitHub stars**.

Torrent — BitTorrent Client in Go Repo

Built a BitTorrent client implementing bencode parsing, tracker communication, the peer wire protocol, piece downloading, and swarm orchestration.

EDUCATION

Vellore Institute of Technology 2024 – 2028

B.Tech in Information Technology

WRITING

Understanding Amazon's DynamoDB Architecture — Partitioning, replication, and consistency in DynamoDB. [Link](#)

The BitTorrent Protocol — Peer-to-peer networking, piece exchange, and protocol-level design. [Link](#)

Understanding Google's Borg — Cluster scheduling, resource allocation, and fault tolerance at scale. [Link](#)

SKILLS

Languages: Go, TypeScript, JavaScript, Rust, SQL

Backend & Systems: Kafka, Node.js, Redis, gRPC, REST APIs, WebRTC, PostgreSQL

Infrastructure: Docker, Kubernetes, AWS, Terraform, Prometheus, Grafana

Frontend: Next.js, React, Tailwind CSS